

BPW1 BLE communication protocol

BPW1

2025/07/25

1.Basic standards of the agreement

- ① The protocol adopts basic unpacking and grouping logic;
- ② The protocol packet is composed of big-end data mode;

2.Bluetooth service and characteristic values

Protocol services					
service function	serve UUID	Notification features	Writeable features	Readable features	
Electricity services	0000 180F-0000-1000-8000-00805F9B34FB	2A19		2A19	
App directive service	81EE A001- E735- 49EC- 8A 11- 7E32CAE1E 14E	A002	A003	A004	

3. Detailed description of the protocol data format

Protocol format (length: 10-512Bytes)									
Protocol identifier	Length of the agreement		Protocol calibration position	protocol command	Version number of the agreement	Protocol subcommand	Command instruction length		Command instruction value
1Byte	2 Byte		1Byte	1Byte	1Byte	1Byte	2 Byte		1~503Byte
	7 bits: reserved	9 bit					7 bits: reserved	9bit	
0xE0	0x0	Lenth	sumcheck	CMD	Version	KEY	0x0	KeyValueLen	KeyValue

Figure data bit description:

- Protocol format: definition of the length range of a single Bluetooth protocol for watches;
- Protocol identifier: This bit represents the Bluetooth protocol identifier of our watch;
- Protocol length: Lenth is the value of protocol data-4;
- Protocol calibration bit: the last 8 bits of the sum of the whole protocol byte data after removing this byte;
- Protocol command: protocol custom command, follow-up details;
- Protocol version number: a flag for subsequent application and protocol upgrade;
- Protocol subcommands: Subcommands based on custom commands, which will be explained in detail later;
- Command instruction length: KeyValueLen is the total number of bits of the command instruction value;
- Command instruction value: detailed data of the protocols custom command subinstruction;

4. Detailed description of data ACK

Protocol ACK (length: 10Bytes)									
ACK flag	Length of the agreement		Protocol calibration position	protocol command	Version number of the agreement	Protocol subcommand	Command instruction length		Error code
1Byte	2Byte		1Byte	1Byte	1Byte	1Byte	2Byte		1Byte
	7 bits: reserved	9bit					7 bits: reserved	9bit	
0x0E	0x0	Lenth	sumcheck	CMD	Version	KEY	0x0	KeyValueLen	KeyValue

ACK error code description:

- 0x00: The protocol is correct
- 0x01: Protocol flag error;
- 0x02: Incorrect protocol length;
- 0x03: Protocol calibration position error;
- 0x04: The length of the command instruction is incorrect;
- 0x05: The protocol version number is incorrect
- 0x06: Protocol command error;
- 0x07: Protocol command instruction error;
- 0x08: Failed to set the sleep time period

The app connects to the watch → The watch responds to the device query command with corresponding data → The app sends the bind command → The watch responds with corresponding data → The app requests the settings list → System time is synchronized → The app sends user information → Real-time exercise is started → Real-time exercise data is transmitted

Device query command (CMD: 0x00KEY: 0x00) (Protocol version number obtained from device reply)

Example of APP issued agreement:

0xE0,0x00,0x06,0xXX,0x00,0x00,0x00,0x00,0x01, 0xxx

KeyValue (1Byte)	
Android	IOS
0x00	0x01

Example of wristband response protocol:

0xE0,0x00,0x0B,0xXX,0x00,0x00,0x00,0x00,0xXX, 0xxx...

KeyValue (2+2+2+2+1+N Byte)					
Gravity chip	Heart rate chip	GPS slug	Other features	Other features	Function expansion
Details of gravity	Heart rate details	GPS detailed information	Other details	Other details	Function details

Details of gravity:

KeyValue (2Byte)							
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	1: chip, 0: no chip
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0: Active sleep 1: Passive sleep	0: Active step counting 1: Passive step counting	0: Movement reservation	0: Movement reservation	0: Movement reservation	0: Movement reservation	1: sleep, 0: no	1: step counting, 0: no

Heart rate details:

KeyValue (2Byte)							
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
0: reserved	0: reserved	0: reserved	0: Reserved	0: reserved	0: reserved	0: reserved	1: chip, 0: no chip
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00: no function, 0x01: only heart rate, 0x02: heart rate + blood pressure, 0x03: heart rate + blood oxygen, 0x04: heart rate + blood pressure + blood oxygen							

GPS detailed information :

KeyValue (2Byte)							
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	0: reserved	1: chip, 0: no chip
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
0x00: no function, 0x01: only positioning							

Other details:

KeyValue (2Byte)							
Bit15	Bit14	Bit13	Bit12	Bit11	Bit10	Bit9	Bit8
Sedentary reminder 1: Support 0: Not supported	Time setting 1: Support 0: Not supported	obligate	obligate	obligate			
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
obligate	obligate	obligate	obligate	Set the time when you raise your hand to display the screen 1: Support 0: Not supported	obligate	obligate	obligate

Other details:

KeyValue (1Byte)							
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
obligate							

Function details:

KeyValue (NByte)								
Expand 1	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1: Do not disturb mode 1: Support 0: Not supported	1: Unit Settings 1: Support 0: Not supported	contextual model 1: Support 0: Not supported	obligate	obligate	Language type data occupies 5Byte 1: Support 0: Not supported (default 4Byte)	obligate	obligate
Expand 2	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	obligate

Extension 3 (Market Dial Type Definition)	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate							
Expand 4	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	Support for uploading device information 1: Support 0: Not supported
Expand 5	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	Automatic blood oxygen measurement 1: Support 0: Not supported
Extension 6	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	obligate
Extension 7	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	The sitting reminder time is accurate to the minute 1: Support 0: Not supported	obligate	obligate	obligate	obligate	obligate	obligate	obligate
Expand 8	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	Drink water to remind 1: Support 0: Not supported	obligate	obligate	obligate	obligate
Extension 9	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	obligate
Expand 10	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate				obligate			
Extension 11	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	obligate	obligate
Expand 12	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	obligate	5610 platform algorithm sleep support 1: Support 0: Not supported	obligate
Expand 13	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	The 5610 sleep algorithm merges REM into the light sleep marker 1: Support 0: Not supported	obligate	obligate	obligate	obligate	obligate	obligate
	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0

Expand 14	561 The sleep algorithm enables the sleep monitoring flag 1: Support 0: Not supported	obligate				obligate	obligate	obligate
Expand 15 obligate	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate		obligate	obligate	obligate
Expand 16	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate		obligate	obligate	obligate		obligate	
Extension 17	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	obligate	obligate	obligate	The heart rate is too high or too low to remind the threshold marker 1: Support 0: Not supported	Low blood oxygen level reminder threshold flag 1: Support 0: Not supported	obligate
Expand 18	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	obligate	obligate	The wake time of the 5610 sleep algorithm is merged into the light sleep flag 1: Support 0: Not supported					

Set command (CMD: 0x02 KEY: multiple-valued)

Raise your hand to light the screen (KEY: 0x00)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x00,0x00,0x01, 0xxx
Early definition:

KeyValue (1Byte)	
close	open
0x00	0x01

On time setting:

KeyValue (5Byte)		
Function switch (1Byte)	Start time (2Byte)	Close time (2Byte)
1: on, 0: off	Number of minutes (Example 8: 30->8*60+30)	Number of minutes (Example 18:30-> 18 * 60 + 30)

Language (KEY: 0x01)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x01,0x00,0x01, 0xxx

KeyValue (1Byte)							
English	simplified Chinese	traditional Chinese	French	Spaish	Polish	Portuguese	Italian
0x00	0x01	0x02	0x03	0x04	0x05	0x06	0x07
German	Dutch	Turkic	Russian	Czech	Farsi	Hungarian	Greek
0x08	0x09	0x0A	0x0B	0x0C	0x0D	0x0E	0x0F
Arabic	Filipino	Malay	Indonesian	Vietnamese	Thai	Burmese	Dard
0x10	0x11	0x12	0x13	0x14	0x15	0x16	0x17
Korean	Japanese	Swedish	Hebrew	the Bay of Bengal	Romanian	Danish	Finnish
0x18	0x19	0x1A	0x1B	0x1C	0x1D	0x1E	0x1F
Ukrainian	Serbo-Croat	Cambodia	Urdu	Slovenian			
0x20	0x21	0x22	0x23	0x24			

Live motion (KEY: 0x02)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x02,0x00,0x01, 0xxx

KeyValue (1Byte)	
close	open
0x00	0x01

Live health (KEY: 0x03)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x03,0x00,0x01, 0xxx

KeyValue (1Byte)	
close	open
0x00	0x01

User information (KEY: 0x04)

Example of APP issued agreement:
0xE0 ,0x00,0x09,0xXX,0x02,0x00,0x04,0x00,0x04, 0xxx

KeyValue (4Byte)				
Gender (1bit)	Age (7 bits)	Height (9bit)	Weight (10 bits)	Retention bit (5 bits)
0: Female, 1: Male	Units (years)	unit (cm)	unit (kg)	--

Target steps (KEY: 0x05)

Example of APP issued agreement:
0xE0 ,0x00,0x09,0xXX,0x02,0x00,0x05,0x00,0x04, 0xxx

Get device Settings information (KEY: 0x06)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x06,0x00,0x01, 0x01
Example of wristband response protocol:
0xE0 ,0x00,0xXX,0xXX,0x02,0x00,0x06,0x00,0xXX, 0xxx

Note: For the new function tag, please pay attention to modify the hyperlink corresponding to the function data item

KeyValue (N Byte)	
Function label (1Byte)	performance data
0x01	User information (4Byte)
0x02	Target steps (4Byte)
0x03	Raise your hand to display the screen + language (5Byte) + real-time movement + real-time health (8Byte)
0x04	Sleep duration (5Byte)
0x05	Wearing mode (1Byte)
0x06	Automatic heart rate measurement (8Byte)
0x07	Sedentary reminder (8Byte or 10Byte) (Function details-Extension 7-Sedentary reminder time is accurate to minute. The marker is 1 and 10Byte)
0x08	12/24 hour system (1Byte)
0x09	Reserve (1Byte)
0x0A	Reserve (20Byte)
0x0B	Screen on time (5Byte)
0x0C	Do Not Disturb mode (5Byte)
0x0D	Unit conversion (2Byte)
0x0E	Scenario mode (1Byte)

0x0F	Reserve (4Byte)
0x10	Screen off duration (1Byte)
0x11	Automatic blood oxygen measurement (8Byte)
0x12	Reserve (1Byte)
0x13	Reserve (1Byte)
0x14	Reserve (1Byte)
0x15	Reserve (8Byte)
0x16	Target calories for the day (4Byte)
0x17	Target distance for the day (4Byte)

Language list description:

KeyValue (5Byte)								
Byte1	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1: English	1: Simplified Chinese	1: Traditional Chinese	1: French	1: Spanish	1) Polish	1: Portuguese	1: Italian
Byte2	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1: German	1: Dutch	1: Turkish	1: Russian	Czech	1: Persian	1: Hungarian	1: Greek
Byte3	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1: Arabic	1: Filipino	1: Malay	1: Indonesian	1) Vietnamese	1: Thai	1: Burmese	1: Hindi
Byte4	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1: Korean	Japanese 1	1: Swedish	1: Hebrew	1: Bengali	1: Romanian	1: Danish	1) Finnish
Byte5	Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
	1) Ukrainian	1: Croatian	1: Khmer	1: Urdu	1: Slovenian	0: reserved	0: reserved	0: reserved

Sleep period (KEY: 0x07)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x07,0x00,0x05, 0xxx

KeyValue (5Byte)		
Function switch (1Byte)	Start time (2Byte)	Close time (2Byte)
1: on, 0: off	Number of minutes (Example 8: 30->8*60+30)	Number of minutes (Example 18:30->18*60+30)

Synchronization system time (KEY: 0x08)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x08,0x00,0x04, 0xxx

KeyValue (4Byte)					
Years (6 bits)	Moon (4 bits)	Day (5 bits)	Time (5 bits)	Bits (6)	seconds (6 bits)
The difference from 2000	1~12	1~31	0~23	0~59	0~59

Restore factory Settings (KEY: 0x09)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x09,0x00,0x01, 0x01

Wearing mode (KEY: 0x0A)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x0A,0x00,0x01, 0xxx

KeyValue (1Byte)	
the left hand	the right hand
0x01	0x02

Automatic heart rate measurement (KEY: 0x0B)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x0B,0x00,0x08, 0xxx

KeyValue (8Byte)				
Function switch (1Byte)	Reserve (1Byte)	Measurement frequency (2Byte)	Start time (2Byte)	Close time (2Byte)
1: on, 0: off	--	10,60,90,120 (unit: min)	Number of minutes (Example 8: 30->8*60+30)	Number of minutes (Example 8: 30->8*60+30)

Sedentary reminder (KEY: 0x0C)

The early warning time is not set to the minute:

Two-way protocol example:

0xE0,0x00,0x06,0xXX,0x02,0x00,0x0C,0x00,0x08, 0xxx

KeyValue (8Byte)						
Do not disturb during lunch break (1Byte)	Function switch (1Byte)	Sedentary threshold (2Byte)	Time spent sitting (1Byte)	Start time (1Byte)	Close time (1Byte)	Redefinition (1Byte)
1: on, 0: off	1: on, 0: off	Prompt to judge the condition value (15: less than 15 steps within the duration)	45,60,75,90,105,120 (minutes) (Add 30 minute option flag extension 12, bit5)	Hour to minute (Example 8: 00->0x08)	Hourly point (Example 8: 00->0x08)	Repeat the definition statement

Set the reminder time to minute:

Two-way protocol example:

0xE0,0x00,0x06,0xXX,0x02,0x00,0x0C,0x00,0x0A, 0xxx

KeyValue (10Byte)						
Do not disturb during lunch break (1Byte)	Function switch (1Byte)	Sedentary threshold (2Byte)	Time spent sitting (1Byte)	Start time (2Byte)	Close time (2Byte)	Redefinition (1Byte)
1: on, 0: off	1: on, 0: off	Prompt to judge the condition value (15: less than 15 steps within the duration)	45,60,75,90,105,120 (minutes) (Add 30 minute option flag extension 12, bit5)	High: Time Low: Divided	High: Time Low: Divided	Repeat definition instructions

Repeat definition explanation:

Redefinition (1Byte)							
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
1: on, 0: off	weekday	Sat	Fri	Thursday	Wednesday	Tuesday	Monday

12/24 hour system (KEY: 0x0D)

Example of APP issued agreement:

0xE0,0x00,0x06,0xXX,0x02,0x00,0x0D,0x00,0x01, 0xxx

KeyValue (1Byte)	
12-hour system	24-hour system
0x00	0x01

Alarm Settings (KEY: 0x10)

Example of APP issued agreement:

0xE0,0x00,0x06,0xXX,0x02,0x00,0x10,0x00,0xXX, 0xxx

KeyValue (2+5*N Byte)										
Quantity (1Byte)		Reminder (1Byte)		Alarm clock attribute (5*N bytes, N=Max)						
Maximum number of alarms (4 bits)	Current number of alarms (4 bits)	Number of bell types (4 bits)	Built-in ringtone count (4 bits)	Time (2Byte)	Repeat type (1Byte)	Bell type (4 bits)	Ringtone index (4 bits)	Alarm switch (2 bits)	Current action (2 bits)	Alarm clock index (4 bits)
Unit: pieces	Unit: pieces	--	--	High: Time Low: Divided	Repeat the definition statement	0: Ring, 1: Vibration, 2: Ring + vibration	0: Device ring 0 1: Device bell 1 ...	0: Off 1: On	0: Delete 1: Settings	Current alarm index: 0~Max-1

Repeat definition explanation:

Redefinition (1Byte)							
Bit7	Bit6	Bit5	Bit4	Bit3	Bit2	Bit1	Bit0
--	weekday	Sat	Fri	Thursday	Wednesday	Tuesday	Monday

Do Not Disturb mode (KEY: 0x11)

Example of APP issued agreement:

0xE0,0x00,0x0E,0xXX,0x02,0x00,0x11,0x00,0x05, 0xxx

KeyValue (5Byte)		
Function switch (1Byte)	Start time (2Byte)	Close time (2Byte)
1: on, 0: off	Number of minutes (Example 8: 30->8*60+30)	Number of minutes (Example 18:30->18*60+30)

Unit conversion (KEY: 0x12)

Example of APP issued agreement:
0xE0 ,0x00,0x0B,0xXX,0x02,0x00,0x12,0x00,0x02, 0xxx

KeyValue (2Byte)	
Unit of distance (1Byte)	Unit of measure (1Byte)
0: meters, 1: feet	0: degrees Celsius, 1: degrees Fahrenheit

Scenario mode (KEY: 0x13)

Example of APP issued agreement:
0xE0 ,0x00,0x0A,0xXX,0x02,0x00,0x13,0x00,0x01, 0xxx

KeyValue (1Byte)	
Value data (1Byte)	Corresponding description
0x00	loudspeaker mute
0x01	diabolo
0x02	shake
0x03	Ring + vibrate

Request device information (KEY: 0x17) (determine whether it is supported according to the function details-> device information upload flag)

0xE0 ,0x0X,0xXX,0xXX,0x02,0x00,0x17,0x0X,0xXX, 0xxx

KeyValue (4+8*N Byte)			
Version number length (1Byte)	Version number (NByte)	Project name length (1Byte)	Project name (NByte)
(N) Byte	Ascll code	(N) Byte	Ascll encoding

The APP requests a long time when the screen is currently on (KEY: 0x1F)

Example of APP request protocol:
0xE0 ,0x00,0x0A,0xXX,0x02,0x00,0x1F,0x00,0x01, 0X00
Example of wristband response protocol:
0xE0 ,0x0X,0xXX,0xXX,0x02,0x00,0x1F,0x0X,0x01, 0xxx
Note: The KeyValue (1Byte) value is the same as the table for setting the screen brightness duration

Set the screen on time (KEY: 0x1C)

Example of APP issued agreement:
0xE0 ,0x00,0x0A,0xXX,0x02,0x00,0x1C,0x00,0x01, 0xXX

KeyValue (1) Byte	
(1Byte) value	Screen time
0x01	Five seconds
0x02	Ten seconds
0x03	15 seconds
0x04	30 seconds
0x05	Sixty seconds
obligate	
0xff	Chang Liang

Automatic blood oxygen measurement (KEY: 0x1D)

Example of APP issued agreement:
0xE0 ,0x00,0x06,0xXX,0x02,0x00,0x1D,0x00,0x08, 0xxx

KeyValue (8Byte)				
Function switch (1Byte)	Reserve (1Byte)	Measurement frequency (2Byte)	Start time (2Byte)	Close time (2Byte)
1: on, 0: off	--	10,60,90,120 (unit: minutes)	Number of minutes (Example 8: 30->8*60+30)	Number of minutes (Example 8: 30->8*60+30)

Set the user name (KEY: 0x20)

Example of APP issued agreement:
0xE0,0x00,0x06,0xFF,0x02,0x00,0x20,0x00,0x01, 0xxx

KeyValue (1+NByte)	
User name length N (unit: bytes)	User name data (UTF-8 encoding)

Water intake reminder (KEY: 0x21)

Value Content description: It is turned off by default. After it is turned on, the default value is: 8:00-22:00; the drinking reminder time is 60 minutes; users can customize it

KeyValue (6Byte)					
(1Byte)	(1Byte)	(1Byte)	(1Byte)	(1Byte)	(1Byte)
noon break	Enable switch	Drink water to remind yourself of the time	Start reminding yourself of the time	End reminder time	Repeat this
No interruptions during lunch break 1: Enable 0: Off	1: Enable the water drinking reminder function 0: Turn off the water reminder function	The minimum is 15 minutes and the maximum is 180 minutes The algorithm is: 15*i	In hours, the range is 0 to 23	In hours, the range is 0~23	0~6bits indicates Monday to Sunday reinvent 1) Repeat 0: No repetition

Target calories (unit: cal) (KEY: 0x2C) (Attached: associated with extended 17, bit5 flag)

Example of APP issued agreement:
0xE0,0x00,0x09,0xFF,0x02,0x00,0x2c,0x00,0x04, 0xxx

Target calorie data format (4Byte)
Calorie data (4Byte)
unit : cal

Target distance (unit: m) (KEY: 0x2D) (Attached: associated with extended 17, bit4 flag)

Example of APP issued agreement:
0xE0,0x00,0x09,0xFF,0x02,0x00,0x2d,0x00,0x04, 0xxx

Target distance data format (4Byte)
Distance data (4Byte)
Unit: m

Synchronize data command (CMD: 0x01 KEY: multi-value) (APP sends synchronous data to device)

Subcommand description table					
KEY	description	KEY	description	KEY	description
0x00	The APP requests synchronization of all data	0x01	The APP requests to synchronize the data of the day	0x02	The device begins to synchronize data
0x03	The device ends the synchronization data				

Note: The device data synchronization process (APP requests data-> the device starts synchronization instructions-> step counting-> sleep-> heart rate-> blood pressure-> blood oxygen-> GPS-> the device ends synchronization instructions) is parsed as follows:

The APP requests to synchronize all data (KEY: 0x00) (the APP sends it to the device)

Example of APP issued agreement:
0xE0,0x00,0x06,0xFF,0x01,0x00,0x00,0x00,0x01, 0x01

The APP requests the synchronization of the data on the same day (KEY: 0x01) (the device replies to the start)

Example of APP issued agreement:

0xE0,0x00,0x06,0xFF,0x01,0x00,0x01,0x00,0x01, 0x01

7.2.3 The device starts to synchronize data (KEY: 0x02) (the device replies to start)

Example of APP issued agreement:

0xE0,0x00,0x06,0xFF,0x01,0x00,0x02,0x00,0x01, 0x01

7.2.4 The device ends the synchronization data (KEY: 0x03) (the device replies the end)

Example of APP issued agreement:

0xE0,0x00,0x06,0xFF,0x01,0x00,0x03,0x00,0x01, 0x01

Health command (CMD: 0x03 KEY: multi-value) (device reply data)

Heart rate (KEY: 0x00)

Example of wristband notification protocol:

0xE0,0x0X,0xFF,0xFF,0x03,0x00,0x00,0x0X,0xFF, 0xxx

KeyValue (4+8*N Byte)			
Data header (4Byte)	Heart rate data (8Byte)	Heart rate data (8Byte)	...
Data header format	Heart rate data format	...	...

Data header format:

Data header format (4Byte)					
Time stamp (2Byte)				Reserve (1Byte)	Number of heart rate data (1Byte)
Reserve (1 bit)	Years (6 bits)	Moon (4 bits)	Day (5 bits)		
--	The difference from 2000	1~12	1~31	--	Unit: pieces

Heart rate data format:

Heart rate data format (8Byte)				
seconds (4Byte)	Heart rate type (1Byte)	Reserve (1Byte)	Reserve (1Byte)	Heart rate value (1Byte)
Number of seconds (Example 7: 30->7*60*60+30*60)	--	--	--	unit : bpm

Blood pressure (KEY: 0x01)

Example of wristband notification protocol:

0xE0,0x0X,0xFF,0xFF,0x03,0x00,0x01,0x0X,0xFF, 0xxx

KeyValue (4+8*N Byte)			
Data header (4Byte)	Blood pressure data (8Byte)	Blood pressure data (8Byte)	...
Data header format	Blood pressure data format	...	...

Data header format:

Data header format (4Byte)					
Time stamp (2Byte)				Reserve (1Byte)	Number of blood pressure data (1Byte)
Reserve (1 bit)	Years (6 bits)	Moon (4 bits)	Day (5 bits)		
--	The difference from 2000	1~12	1~31	--	Unit: pieces

Blood pressure data format:

Blood pressure data format (8Byte)				
seconds (4Byte)	Blood pressure type (1Byte)	Reserve (1Byte)	High value (1Byte)	Low value (1Byte)
Number of seconds (Example 7: 30->7*60*60+30*60)	--	--	unit : mmHg	unit : mmHg

Blood oxygen (KEY: 0x02)

Example of wristband notification protocol:

0xE0,0x0X,0xFF,0xFF,0x03,0x00,0x02,0x0X,0xFF, 0xxx

KeyValue (4+8*N Byte)			
Data header (4Byte)	Oxygen data (8Byte)	Oxygen data (8Byte)	...
Data header format	Blood oxygen data format	...	...

Data header format:

Data header format (4Byte)					
Time stamp (2Byte)				Reserve (1Byte)	Number of blood oxygen data (1Byte)
Reserve (1bit)	Years (6 bits)	Moon (4 bits)	Day (5 bits)		

--	The difference from 2000	1~12	1~31	--	Unit: pieces
----	--------------------------	------	------	----	--------------

Blood oxygen data format:

Blood oxygen data format (8Byte)				
seconds (4Byte)	Type of blood oxygen (1Byte)	Reserve (1Byte)	Reserve (1Byte)	Blood oxygen value (1Byte)
Number of seconds (Example 7: 30->7*60*60+30*60)	--	--	--	unit : %

APP requests the threshold for high and low heart rate alerts (CMD: 0x03 KEY: 0x08) (Attached: associated with extended 17, bit2 flag)

Example of APP issued agreement:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x08,0x00,0x01, 0x01

Example of device reply protocol:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x08,0x00,0x10, 0xxx

KeyValue (10Byte)									
Heart rate threshold (5Byte)					Low heart rate threshold (5Byte)				
Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte	Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte
0: off; 1: on	Example: 110/120/130/140/150 times/min, step 10				0: off; 1: on	Example: 40/45/50 times per minute, step 5			

The device reports the threshold for high and low heart rate alerts (CMD: 0x03 KEY: 0x09) (Attached: associated with extended 17, bit2 flag)

Device notification protocol example:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x09,0x00,0x10, 0xxx

Example of APP reply protocol:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x09,0x00,0x10 ,0xxx

KeyValue (10Byte)									
Heart rate threshold (5Byte)					Low heart rate threshold (5Byte)				
Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte	Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte
0: off; 1: on	Example: 110/120/130/140/150 times/min, step 10				0: off; 1: on	Example: 40/45/50 times per minute, step 5			

Set the threshold of heart rate warning too high or too low in the APP (CMD: 0x03 KEY: 0x0A) (Attached: associated with extended 17, bit2 flag)

Example of APP issued agreement:

0xE0,0x00,0x0E,0xXX,0x03,0x00,0x0A,0x00,0x04, 0xxx

KeyValue (4Byte)			
Heart rate threshold (2Byte)		Low heart rate threshold (2Byte)	
Switch 1Byte (0: off; 1: on)	The current set value is 1Byte	Switch 1Byte (0: off; 1: on)	The current set value is 1Byte

APP requests the hypoxia alert threshold (CMD: 0x03 KEY: 0x0B) (Attached: associated with extended 17, bit1 flag)

Example of APP issued agreement:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x0B,0x00,0x01, 0x01

Example of device reply protocol:

0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x0B,0x00,0x05, 0xxx

KeyValue (5Byte)				
Oxygen hypoxia threshold (5Byte)				
Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte
0: off; 1: on	Example: 75/80/85/90, step 5			

The device reports the hypoxia warning threshold (CMD: 0x03 KEY: 0x0C) (Attached: associated with extended 17, bit1 flag)

Device notification protocol example:
0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x0C,0x00,0x05, 0xxx
Example of APP reply protocol:
0xE0,0x0X,0xXX,0xXX,0x03,0x00,0x0C,0x00,0x05 ,0xxx

KeyValue (5Byte)				
Oxygen hypoxia threshold (5Byte)				
Switch 1Byte	The maximum value is 1Byte	Minimum value is 1Byte	The step value is 1Byte	The current value is 1Byte
0: off; 1: on	Example: 75/80/85/90, step 5			

Set the hypoxia low alert threshold of the APP (CMD: 0x03 KEY: 0x0D) (Attached: associated with extended 17, bit1 flag)

Example of APP issued agreement:
0xE0,0x00,0x0E,0xXX,0x03,0x00,0x0D,0x00,0x01, 0xxx

KeyValue (2Byte)	
Oxygen hypoxia threshold (2Byte)	
Switch 1Byte (0: off; 1: on)	The current set value is 1Byte

Motion command (CMD: 0x04 KEY: multiple-valued)

Total step data on the day (KEY: 0x00)

Example of wristband notification protocol:
0xE0,0x00,0x11,0xXX,0x04,0x00,0x00,0x00,x0C, 0xxx

KeyValue (12Byte)		
Number of steps (4Byte)	Distance (4Byte)	Calories (4Byte)
Step size	Unit: m	unit : Kcal

Complete step data of device (KEY: 0x01)

Example of wristband notification protocol:
0xE0,0x00,0x11,0xXX,0x04,0x00,0x01,0x01,x24, 0xxx

KeyValue (4+12*24 Byte)			
Data header (4Byte)	Time-sharing step data (12Byte)	...	Time-sharing step data (12Byte)
Data header format	Time-sharing step data format	...	...

Data header format:

Data header format (4Byte)					
Time stamp (2Byte)				Reserve (1Byte)	Number of steps counted per time (1Byte)
Reserve (1 bit)	Years (6 bits)	Moon (4 bits)	Day (5 bits)		
--	The difference from 2000	1~12	1~31	--	Fixed value: 24

5610 Platform algorithm historical sleep result data format (KEY: 0x10)

Example of wristband notification protocol:
0xE0,0xXX,0xXX,0xXX,0x04,0xXX,0x10,0x0X,xXX, 0xxx

KeyValue (4+5*N Byte+4*N Byte)				
4 Byte	5 Byte	...	4 Byte	...
data head	Sleep results data	...	Data on the results of naps	...
Data header format	Sleep result data format	...	Data format for nap results	...

Data header format:

Data header format (4Byte)					
Time stamp (2Byte)				1 Byte	1 Byte
1 bit	6 bit	4 bit	5 bit		
obligate	year	moon	sun	Number of sleep result data	Number of data results from naps

--	The difference from 2000	1~12	1~31	Unit: pieces	Unit: pieces
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Sleep result data format:

Sleep result data format (5Byte)		
2 Byte	2 Byte	1 Byte
starting time	terminal time	sleep state
Number of minutes (Example 18:30->18*60+30)	Number of minutes (Example 18:30-> 18 * 60 + 30)	0-not asleep, 1-light sleep, 2-deep sleep, 3-eye movement, 4-nap, 5-temporary wakefulness

Data format of naps results:

Data format for nap result (4Byte)	
2 Byte	2 Byte
starting time	terminal time
Number of minutes (Example 18:30-> 18 * 60 + 30)	Number of minutes (Example 18:30->18*60+30)

APP control device command (CMD: 0x06 KEY: multiple-valued)

Request device heart rate data (KEY: 0x04) (data format Health Command-Heart Rate)

Example of APP issued agreement:
0xE0,0x00,0x0A,0xXX,0x06,0x00,0x04,0x00,0x01, 0x01

Request device blood pressure data (KEY: 0x05) (data format Health Command-Blood Pressure)

Example of APP issued agreement:
0xE0,0x00,0x0A,0xXX,0x06,0x00,0x05,0x00,0x01, 0x01

Request device blood oxygen data (KEY: 0x06) (data format Health Command-Blood Oxygen)

Example of APP issued agreement:
0xE0,0x00,0x0A,0xXX,0x06,0x00,0x06,0x00,0x01, 0x01

Bind device (CMD: 0x08 KEY: 0x00)

Example of APP issued agreement:
0xE0,0x00,0x06,0xXX,0x08,0x00,0x00,0x00,0x08, 0xxx
0xxx: 8-byte binding code
Example of wristband response protocol:

0xE0,0x00,0xXX,0xXX,0x08,0x00,0x00,0x00,0x01, 0xxx

KeyValue (1Byte)		
The device is initially bound for reply	The device returns to the APP	The device refuses to bind the reply
0x01	0x02	0x03

Unbind device (CMD: 0x08 KEY: 0x01)

Example of APP issued agreement:
0xE0,0x00,0x0A,0xXX,0x08,0x00,0x01,0x00,0x01, 0x01